

AURA FRAUD INVESTIGATION REPORT

AI-Assisted Transaction Investigation

1. Transaction Overview

Transaction ID	98e3dcf98101146a577f85a34e58feec
Risk Band	MEDIUM
Risk Score	38.82
Fraud Probability	0.00%
Transaction Amount	7.27
Investigation Flag	False
Recommended Action	SEND TO REVIEW QUEUE

2. Executive Summary

Transaction 98e3dcf98101146a577f85a34e58feec has been classified as MEDIUM risk with an AURA risk score of 38.82. The transaction should be reviewed through the appropriate investigation queue.

3. Risk Interpretation

- The AURA risk engine has not flagged this transaction for investigation.
- The machine-learning fraud probability is 0.00%.
- The transaction amount is 7.27.

4. Investigation Evidence

Evidence Type	Description
investigation_reason	Previously unseen merchant for this customer
investigation_reason	Previously unseen transaction category
investigation_reason	No prior transaction history available

5. Top Risk Factors

Feature	Observed Value	SHAP Value
customer_prev_count	0	0.03396016096577136
is_new_category	1	0.0176966167431728
category_seen_before	0	0.012207135396711514
is_new_merchant	1	0.0028178099566009635

Feature	Observed Value	SHAP Value
merchant_seen_before	0	0.0002803552647720063

6. Graph Analysis

Metric	Value
Graph Risk Score	14.507072579947925
Graph Risk Band	LOW
Transaction Count	1518
Fraud Transaction Count	9
Fraud Ratio	0.005928853754940711

Graph Reasons:

- Customer is connected to previously identified fraudulent transaction(s)

7. AI Investigation Analysis

AI Assessment

- The AURA engine assigns a *Medium* risk band (score = 38.82) with a *0 %* fraud probability.
- The transaction is flagged for review but not for immediate escalation.
- The deterministic investigation confirms that the model did not trigger an alert, yet it notes three investigation reasons (unseen merchant, unseen category, no prior history).

Key Fraud Indicators (Evidence)

Indicator	Evidence	Weight (SHAP)
New merchant	`is_new_merchant = 1`, `merchant_seen_before = 0`	0.0028
New category	`is_new_category = 1`, `category_seen_before = 0`	0.0177
No prior customer history	`customer_prev_count = 0`	0.03396
Novelty risk	100 % (max)	-
Graph connection	Connected to known fraud graph; `graph_risk_band = LOW` but `graph_risk_score = 14.51`	-

**Note:* The anomaly score is very low (0.028) and the amount deviation metrics are unavailable due to the absence of prior transactions.

Evidence Interpretation

Observation	Interpretation
No transaction history	The customer has zero prior records; therefore all behavioural baselines are undefined, inflating novelty metrics.
Unseen merchant & category	New merchant and category are behavioural deviations; per the <i>*merchant_fraud*</i> and <i>*behavioural_risk*</i> knowledge, such factors alone do **not** prove fraud but are signals requiring context.
Low anomaly/amount metrics	The transaction amount (7.27 USD) is modest; absence of a deviation ratio or velocity data further reduces suspicion.
Graph linkage	The customer node is connected to a fraud-identified sub-graph, which elevates graph risk slightly; however, the graph risk band remains <i>*LOW*</i> and does not override the medium overall score.
Risk score vs fraud probability	The risk engine separates a numeric risk score from a probabilistic fraud prediction. A zero fraud probability indicates the model sees no convincing evidence of fraud given available data.

Investigation Considerations

1. Missing Historical Data - The lack of prior amounts, frequency, and location limits behavioural comparison.
2. Merchant Legitimacy - No prior interaction raises questions but is insufficient for denial.
3. Geographical Context - Customer-merchant distance (~128 km) is noted; verify that the merchant's location aligns with the customer's expected travel patterns.
4. Graph Context - Although graph risk is low, the connection to known fraud nodes warrants a closer look at the surrounding network (e.g., shared IPs, device IDs).
5. Cardholder Confirmation - The transaction is recent; verifying the cardholder's intent can rule out misuse.

Recommended Next Steps

1. Obtain Cardholder Confirmation - Contact the customer to confirm the purchase (e.g., via SMS or call).
2. Verify Merchant Identity - Cross-check the merchant name, address, and business registration to ensure it is legitimate.
3. Assess Transaction Location - Confirm that the transaction point of sale aligns with the customer's known location or travel itinerary.
4. Examine Connected Accounts - Review the graph for other linked accounts or devices that may share fraudulent patterns.
5. Flag for Manual Review - Place the transaction in the review queue with the noted investigation reasons to allow analysts to apply additional context.

6. Document Findings - Record any new evidence or lack thereof to refine future model training and risk scoring.

All steps should preserve the original risk score, fraud probability, and risk band as provided.

8. Investigator Conclusion

The transaction should be reviewed for additional evidence before determining whether escalation is required.